

Create-a-Recipe Ratio Challenge

Unit 3: Culinary Academy · Lesson 3-1 · Flagship · EduWonderLab

UNIT 3 · LESSON 3-1 ★ FLAGSHIP · TEACHER NOTES

Standard 6.RP.1

FORMAT Design + trade challenge	TIME 30–35 min
GROUPING Teams of 3–4	TOPIC Understand Ratios (Flagship)
STANDARD 6.RP.1	PREP LEVEL Medium — print blank recipe cards and ingredient icons

Learning goal *I can design a recipe with ratio constraints and find all the ratios in a recipe.*

Teacher setup

- Print blank cards and icons per team. Give each team one constraint card.
- Teams design a recipe that meets the constraint, then trade with another team.
- Receiving teams list every ratio they can find.

Materials

Blank recipe cards, Ingredient icons, Ratio checklist, Pencil.

Differentiation & language support

- Icons let ESOL students build recipes without heavy reading.
- Checklist: part-to-part ratios, part-to-whole ratios, total amount (SPED).
- Sentence stem: “One ratio in this recipe is ____ : ____, which compares ____.”

Key vocabulary (English / Español):

Term	Español	What it means
constraint	restricción	a rule the design must follow
equivalent ratio	razón equivalente	a ratio with the same value

Quick teacher check: *Check that scaled recipes keep equivalent ratios (Card 6 thinking).*

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Name: _____

Date: _____

Class: _____

Your goal: *I can design a recipe with ratio constraints and find all the ratios in a recipe.*

What you need

Blank recipe cards, Ingredient icons, Ratio checklist.

How to play

1. Read your constraint card and design a recipe that follows it.
2. Write the amounts using ingredient icons or labels.
3. Trade recipes with another team.
4. List every part-to-part and part-to-whole ratio you can find.

Constraint Cards

Design a recipe that follows your constraint, then find every ratio in a partner team's recipe.

1. Constraint: Use a 2:3 ratio of two ingredients. Example?	2. Constraint: One ingredient is twice another (2:1). Example?
3. Constraint: Total of exactly 10 cups. List a part-to-whole ratio.	4. Constraint: A 3:2:1 ratio of three ingredients. Example?
5. Find all part-to-part ratios in: 2 cups rice, 4 cups water, 1 cup beans.	6. Challenge: Write two equivalent ratios for a recipe of 3 cups mix : 6 cups water.

Recipe design + ratios found:

Explain your thinking

Explain how one recipe can contain several different ratios at the same time.

Sentence frame: *This recipe has the ratio ___ : ___ when I compare ___ to ___.*

★ **Challenge:** *Scale your recipe to feed twice as many people. Show that every ratio stays the same.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Constraint: Use a 2:3 ratio of two ingredients. Example? Answer: e.g., 2 cups oats : 3 cups flour (any multiple works)	2. Constraint: One ingredient is twice another (2:1). Example? Answer: e.g., 4 cups water : 2 cups mix
3. Constraint: Total of exactly 10 cups. List a part-to-whole ratio. Answer: e.g., 4 cups sugar : 10 cups total → 4:10	4. Constraint: A 3:2:1 ratio of three ingredients. Example? Answer: e.g., 3 flour : 2 sugar : 1 butter
5. Find all part-to-part ratios in: 2 cups rice, 4 cups water, 1 cup beans. Answer: 2:4 (rice:water), 2:1 (rice:beans), 4:1 (water:beans)	6. Challenge: Write two equivalent ratios for a recipe of 3 cups mix : 6 cups water. Answer: e.g., 1:2 and 6:12 (any equivalent to 3:6)

Teaching notes & look-fors

- A recipe with 3 ingredients has 3 part-to-part comparisons.
- Scaling multiplies every amount by the same number, keeping ratios equivalent.